

16.3 Balanced two-way layouts via orthogonal projections

With $m \geq 2$ observations in every cell, the replicates supply an estimate of σ^2 that is valid whatever the cell means are, and interaction can be tested without any assumption about its form. This section gives the complete analysis of the balanced two-way layout: the orthogonal decomposition, exact tests, contrasts, and simultaneous inference. Exercise (C1) asked for the five projections, and Theorem 16.1.2(d) has already met four of them.

16.3.1 The decomposition

The model is the cell-means model

$$E(y_{ijk}) = \mu_{ij}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}, \quad i \leq a, j \leq b, k \leq m, \quad (16.3.1)$$

written in the effects form of Definition 16.2.1 as $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$, with equal-weight effects. Its mean space $\mathcal{C}(\mathbf{X})$ is the set of vectors that are constant within cells, of dimension ab (Proposition 8.6.2). With the ordering and notation of Section 16.1, define

$$\begin{aligned} \mathbf{P}_0 &= \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_A &= \mathbf{C}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_B &= \bar{\mathbf{J}}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m, \\ \mathbf{P}_{AB} &= \mathbf{C}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_E &= \mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{C}_m. \end{aligned} \quad (16.3.2)$$

Theorem 16.3.1 — The balanced two-way layout

Assume Eq. 16.3.1 with $a, b \geq 2$ and $m \geq 2$, and let $\alpha_i, \beta_j, \gamma_{ij}$ be the equal-weight effects of Definition 16.2.1.

- (a) The five matrices Eq. 16.3.2 are mutually orthogonal projections that sum to $\mathbf{I}$, with ranks 1, $a - 1$, $b - 1$, $(a - 1)(b - 1)$ and $ab(m - 1)$. The projection onto $\mathcal{C}(\mathbf{X})$ is $\mathbf{M} = \mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B + \mathbf{P}_{AB}$, and that onto the additive model space is $\mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B$. So $\mathcal{C}(\mathbf{P}_{AB})$ is the orthogonal complement of the additive model within the cell-means model.
- (b) The projections act by

$$\begin{aligned} (\mathbf{P}_{AY})_{ijk} &= \bar{y}_{i..} - \bar{y}_{...}, \\ (\mathbf{P}_{ABY})_{ijk} &= \bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}, \\ (\mathbf{P}_{EY})_{ijk} &= y_{ijk} - \bar{y}_{ij.}, \end{aligned}$$

and symmetrically for $\mathbf{P}_B$. The sums of squares are

$$\begin{aligned} \text{SS}_A &= bm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2, \\ \text{SS}_B &= am \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2, \\ \text{SS}_{AB} &= m \sum_{i,j} (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2, \\ \text{SSE} &= \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij.})^2. \end{aligned}$$

(c) The mean squares have expectations

$$\begin{aligned} \text{E MS}_A &= \sigma^2 + \frac{bm}{a-1} \sum_i \alpha_i^2, \\ \text{E MS}_B &= \sigma^2 + \frac{am}{b-1} \sum_j \beta_j^2, \\ \text{E MS}_{AB} &= \sigma^2 + \frac{m}{(a-1)(b-1)} \sum_{i,j} \gamma_{ij}^2, \\ \text{E MSE} &= \sigma^2. \end{aligned}$$

(d) Under normal errors the four sums of squares are independent, $\text{SSE}/\sigma^2 \sim \chi^2(ab(m-1))$, and

$$\begin{aligned} F_A &= \frac{\text{MS}_A}{\text{MSE}} \sim F\left(a-1, ab(m-1), \frac{bm}{\sigma^2} \sum_i \alpha_i^2\right), \\ F_{AB} &= \frac{\text{MS}_{AB}}{\text{MSE}} \sim F\left((a-1)(b-1), ab(m-1), \frac{m}{\sigma^2} \sum_{i,j} \gamma_{ij}^2\right), \end{aligned}$$

and likewise for F_B . So F_{AB} is an exact test of additivity, F_A of $\bar{\mu}_{1.} = \cdots = \bar{\mu}_{a.}$, and F_B of $\bar{\mu}_{.1} = \cdots = \bar{\mu}_{.b}$.

(e) SS_A is the sum of squares of the hypothesis $\alpha_1 = \cdots = \alpha_a = 0$ in the cell-means model. It is also the sequential sum of squares of A whether A is entered before or after B , and it equals the Type II and Type III sums of squares of A . The same holds for B , and SS_{AB} is the sequential, Type II and Type III sum of squares of the interaction.

Proof.

- (a) Orthogonality, ranks and the sum $\mathbf{I}$ follow from Lemma 16.1.1, as in the proof of Theorem 16.1.2(d). The sum of the first four is $(\bar{\mathbf{J}}_a + \mathbf{C}_a) \otimes (\bar{\mathbf{J}}_b + \mathbf{C}_b) \otimes \bar{\mathbf{J}}_m = \mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m$. This matrix replaces each observation by its cell mean, so it is the projection onto the vectors that are constant within cells, which is $\mathcal{C}(\mathbf{X})$. The additive projection is Theorem 16.1.2(a).
- (b) Expand each Kronecker product with $\mathbf{C} = \mathbf{I} - \bar{\mathbf{J}}$ and read off the means from the table in Section 16.1. For example,

$$\begin{aligned} \mathbf{P}_{AB} &= \mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m - \mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m \\ &\quad - \bar{\mathbf{J}}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m + \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m. \end{aligned}$$

Each sum of squares is a sum over the n positions of a quantity that is constant over m , bm or am of them.

- (c) Apply (b) to the mean vector $\boldsymbol{\mu}$, whose entries are μ_{ij} . Then $\mathbf{P}_A \boldsymbol{\mu}$ has entries $\bar{\mu}_{i.} - \bar{\mu}_{..} = \alpha_i$, so $\|\mathbf{P}_A \boldsymbol{\mu}\|^2 = bm \sum_i \alpha_i^2$. Likewise $\|\mathbf{P}_{AB} \boldsymbol{\mu}\|^2 = m \sum_{i,j} \gamma_{ij}^2$, and $\mathbf{P}_E \boldsymbol{\mu} = \mathbf{0}$. The four subspaces $\mathcal{C}(\mathbf{P}_0), \dots, \mathcal{C}(\mathbf{P}_{AB})$ form an analysis of variance decomposition of $\mathcal{C}(\mathbf{X})$, so Theorem 9.6.1 gives the expectations.
- (d) This is Corollary 9.6.2 with the noncentralities of (c). The noncentrality of F_A vanishes iff every $\alpha_i = 0$, that is, iff the row means $\bar{\mu}_{i.}$ are equal. The noncentrality of F_{AB} vanishes iff every $\gamma_{ij} = 0$, which is additivity by Proposition 16.2.1(vi).
- (e) Each α_i is a linear function of the cell means, so the hypothesis is testable. The mean vectors it allows are the $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ with $\mathbf{P}_A \boldsymbol{\mu} = \mathbf{0}$. Since $\boldsymbol{\mu} = \mathbf{P}_0 \boldsymbol{\mu} + \mathbf{P}_A \boldsymbol{\mu} + \mathbf{P}_B \boldsymbol{\mu} + \mathbf{P}_{AB} \boldsymbol{\mu}$, these form $\mathcal{S}_0 = \mathcal{C}(\mathbf{P}_0) \oplus \mathcal{C}(\mathbf{P}_B) \oplus \mathcal{C}(\mathbf{P}_{AB})$. By Theorem 11.3.2(b) the sum of squares of the hypothesis is $\|(\mathbf{M} - \mathbf{P}_{S_0}) \mathbf{y}\|^2 = \|\mathbf{P}_{AY}\|^2$. The rest is [Chapter 9](#): equal counts are proportional (Theorem 9.4.3), so Theorem 9.4.1 applies, and Exercise (B1) shows that the sequential sums of squares of A in either order and its Type II and Type III sums of squares all equal $\|\mathbf{P}_{AY}\|^2$. The interaction enters last in every order, so all its sums of squares are $\|(\mathbf{M} - \mathbf{P}_0 - \mathbf{P}_A - \mathbf{P}_B) \mathbf{y}\|^2 = \|\mathbf{P}_{ABY}\|^2$.

■

The theorem is summarized in the analysis of variance table:

Source	df	Sum of squares	E(MS)
A	$a - 1$	$\ \mathbf{P}_{AY}\ ^2$	$\sigma^2 + bm \sum_i \alpha_i^2 / (a - 1)$
B	$b - 1$	$\ \mathbf{P}_{BY}\ ^2$	$\sigma^2 + am \sum_j \beta_j^2 / (b - 1)$
AB	$(a - 1)(b - 1)$	$\ \mathbf{P}_{ABY}\ ^2$	$\sigma^2 + m \sum_{i,j} \gamma_{ij}^2 / ((a - 1)(b - 1))$
error	$ab(m - 1)$	$\ \mathbf{P}_{EY}\ ^2$	σ^2
corrected total	$abm - 1$	$\ (\mathbf{I} - \mathbf{P}_0) \mathbf{y}\ ^2$	

The test for A is valid with or without interaction, because $\mathcal{C}(\mathbf{P}_{AB}) \perp \mathcal{C}(\mathbf{P}_A)$. What the interaction changes is the *meaning* of H_A : equality of equal-weight row means is one hypothesis among many when the simple effects differ ([Section 16.2](#)). Part (e) is special to balance. With unequal cell counts the sequential, Type II and Type III sums of squares of A test hypotheses with different weights (Theorem 9.3.2), the subject of [Chapter 17](#).

16.3.2 Contrasts and their sums of squares

The F tests say whether a term matters. Contrasts say how. A **main-effect contrast** in A is $\psi = \sum_i c_i \bar{\mu}_{i.} = \sum_i c_i \alpha_i$ with $\sum_i c_i = 0$. An **interaction contrast** is $\sum_{i,j} d_{ij} \mu_{ij} = \sum_{i,j} d_{ij} \gamma_{ij}$, with (d_{ij}) having zero row and column sums (Proposition 16.2.1). Their least squares estimators replace means by sample means. Since $\bar{y}_{i.}$ averages bm independent observations and $\bar{y}_{ij.}$ averages m ,

$$\text{Var}\left(\sum_i c_i \bar{y}_{i.}\right) = \frac{\sigma^2}{bm} \sum_i c_i^2, \quad \text{Var}\left(\sum_{i,j} d_{ij} \bar{y}_{ij.}\right) = \frac{\sigma^2}{m} \sum_{i,j} d_{ij}^2.$$

The sums of squares of the corresponding single-degree-of-freedom hypotheses are the squared estimates divided by these variances without the σ^2 (Theorem 9.4.4), as for a one-way contrast

(Proposition 15.2.1):

$$SS(\mathbf{c}) = \frac{(\sum_i c_i \bar{y}_{i..})^2}{\sum_i c_i^2 / (bm)}, \quad SS(\mathbf{d}) = \frac{(\sum_{i,j} d_{ij} \bar{y}_{ij.})^2}{\sum_{i,j} d_{ij}^2 / m}. \quad (16.3.3)$$

For main effects, a complete set of $a - 1$ mutually orthogonal contrasts ($\sum_i c_i c'_i = 0$) splits SS_A into $a - 1$ independent pieces. This is the one-way result Theorem 15.4.1 applied to the row means, which are one-way means with bm observations each. For interaction, orthogonal contrasts are most easily built as products.

Proposition 16.3.2 — Product contrasts

Let $\mathbf{c}^{(1)}, \dots, \mathbf{c}^{(a-1)}$ be mutually orthogonal contrast vectors in $\mathbb{R}^a$, and $\mathbf{e}^{(1)}, \dots, \mathbf{e}^{(b-1)}$ mutually orthogonal contrast vectors in $\mathbb{R}^b$. Then the $(a - 1)(b - 1)$ tables $\mathbf{d}^{(rs)} = \mathbf{c}^{(r)} \mathbf{e}^{(s)\top}$, with entries $c_i^{(r)} e_j^{(s)}$, are coefficient tables of mutually orthogonal interaction contrasts, and

$$\sum_{r=1}^{a-1} \sum_{s=1}^{b-1} SS(\mathbf{d}^{(rs)}) = SS_{AB}.$$

Under normality the $(a - 1)(b - 1)$ sums of squares are independent of each other and of SSE.

Proof. The row sums of $\mathbf{d}^{(rs)}$ are $c_i^{(r)} \sum_j e_j^{(s)} = 0$ and the column sums vanish likewise. Let $\mathbf{v}_{rs} = \mathbf{c}^{(r)} \otimes \mathbf{e}^{(s)} \otimes \mathbf{1}_m \in \mathbb{R}^n$. Since $\mathbf{C}_a \mathbf{c}^{(r)} = \mathbf{c}^{(r)}$ and $\mathbf{C}_b \mathbf{e}^{(s)} = \mathbf{e}^{(s)}$, the mixed-product rule gives $\mathbf{P}_{AB} \mathbf{v}_{rs} = \mathbf{v}_{rs}$, so $\mathbf{v}_{rs} \in \mathcal{C}(\mathbf{P}_{AB})$. Also $\mathbf{v}_{rs}^\top \mathbf{v}_{r's'} = m (\mathbf{c}^{(r)\top} \mathbf{c}^{(r')})(\mathbf{e}^{(s)\top} \mathbf{e}^{(s')})$, which is zero unless $r = r'$ and $s = s'$. So the $\mathbf{v}_{rs}$ are $(a - 1)(b - 1)$ nonzero orthogonal vectors in a space of that dimension, an orthogonal basis of $\mathcal{C}(\mathbf{P}_{AB})$. Hence $\|\mathbf{P}_{AB} \mathbf{y}\|^2 = \sum_{r,s} (\mathbf{v}_{rs}^\top \mathbf{y})^2 / \|\mathbf{v}_{rs}\|^2$. Finally $\mathbf{v}_{rs}^\top \mathbf{y} = m \sum_{i,j} d_{ij}^{(rs)} \bar{y}_{ij.}$ and $\|\mathbf{v}_{rs}\|^2 = m \sum_{i,j} (d_{ij}^{(rs)})^2$, so each term is $SS(\mathbf{d}^{(rs)})$ of Eq. 16.3.3. Independence follows from Theorem 4.4.4 applied to the projections onto the lines $\text{span}(\mathbf{v}_{rs})$ and $\mathbf{P}_E$. ■

Product contrasts suit factors with structure. If $\mathbf{c}$ compares a control with the treatments and $\mathbf{e}$ is a linear trend in a quantitative factor, then $\mathbf{c}\mathbf{e}^\top$ asks whether the trend differs between them. The cell-means model of a balanced layout with quantitative levels w_i and z_j is the same as the polynomial model with all terms $w^r z^s$, $r < a$, $s < b$ (Exercise (B3)), and the product contrasts of orthogonal polynomials give its sequential sums of squares.

16.3.3 Simultaneous inference

The tests and contrasts above come in families, and the methods of Chapter 13 apply directly.

- **Tukey for the levels of one factor.** The row means $\bar{y}_{i..}$ are independent, each with variance $\sigma^2 / (bm)$. They are functions of $(\mathbf{I} - \mathbf{P}_E) \mathbf{Y}$, hence independent of MSE, with or without interaction (Exercise (B2)). So Theorem 13.3.2 applies with $g = a$, multiplier $q_\alpha(a, ab(m - 1))$ and standard error $s / \sqrt{bm}$.

- **Scheffé for interaction contrasts.** Taking $\mathcal{V} = \mathcal{C}(\mathbf{P}_{AB})$ in Theorem 13.2.2 gives intervals for all interaction contrasts at once, with multiplier $\sqrt{(a-1)(b-1)F_\alpha((a-1)(b-1), ab(m-1))}$. By Corollary 13.2.3, some interaction contrast is Scheffé-significant iff the F_{AB} test rejects. This is the honest way to report a contrast suggested by the interaction plot. The most extreme contrast always accounts for the whole interaction sum of squares ([Exercise \(C1\)](#)); the question is whether it is interpretable.
- **Bonferroni or Holm for planned families** (Theorem 13.4.1, Theorem 13.4.5), such as the comparison of two levels of A separately at each level of B .

Example. (Analysis of the bread trial)

For the data of Example (Flour and proofing temperature), the decomposition of [Theorem 16.3.1](#) gives

Source	df	Sum of squares	Mean square	F	p
flour	2	1627629	813814	255.74	6.6×10^{-17}
temperature	3	140499	46833	14.72	1.2×10^{-5}
interaction	6	98947	16491	5.18	1.5×10^{-3}
error	24	76373	3182		

so $s = 56.4$ ml, and the interaction is clearly present. The script confirms that statsmodels reports the same sums of squares for Types I, II and III, as part (e) says.

Where is the interaction? Take the flour contrasts $(2, -1, -1)$ (white against the other two) and $(0, 1, -1)$ (wholemeal against rye), and the orthogonal polynomial contrasts $(-3, -1, 1, 3)$, $(1, -1, -1, 1)$, $(-1, 3, -3, 1)$ for the equally spaced temperatures. The six products split SS_{AB} as in [Proposition 16.3.2](#):

	linear	quadratic	cubic
white vs others	52152 ($F = 16.39$)	3134 ($F = 0.98$)	6011 ($F = 1.89$)
wholemeal vs rye	8118 ($F = 2.55$)	18872 ($F = 5.93$)	10660 ($F = 3.35$)

The six add up to 98947. The largest piece, a fraction 0.527 of the interaction, says that white flour responds to temperature more steeply than the others. The second is the wholemeal peak at 32 °C. Does the white-by-linear component survive the most generous correction? Its estimate is 1444.3. The temperature contrast has $\sum_j e_j t_j = -72 - 28 + 32 + 108 = 40$, and the flour contrast is twice “white minus the average of the others”, so the estimate divided by 80 is the difference between white flour’s slope and the average slope of the other two: 18.05 ml per °C. The Scheffé multiplier for interaction contrasts is 3.879, and the simultaneous interval for this slope difference is $[0.75, 35.35]$ ml per °C. It excludes zero even though Scheffé’s correction allows for every contrast the plot might have suggested. The temperature main effect is almost entirely linear, with sequential sums of squares 134098, 6373 and 28 for the linear, quadratic and cubic components.

Comparing the flours. The equal-weight flour means are 2074.8, 1727.1 and 1565.2 ml. Tukey’s multiplier $q_{0.05}(3, 24) = 3.532$ with standard error $s/\sqrt{12}$ gives half-width 57.5 ml. White minus wholemeal is 347.7, with interval $[290.2, 405.3]$. White minus rye blend is 509.7 $[452.2, 567.2]$, and wholemeal minus rye blend is 161.9 $[104.4, 219.4]$. Because of the interaction, these average over the four temperatures the trial happened to use. The simple effects of white against wholemeal, with Bonferroni intervals for the four temperatures (multiplier $t_{24, 0.05/8} = 2.700$, half-width 124.4), are 250.7 $[126.3, 375.0]$, 376.7 $[252.3, 501.0]$, 302.3 $[178.0, 426.7]$ and 461.3 $[337.0, 585.7]$. All four lie above zero, so with simultaneous confidence 95% white flour is better at every temperature, and hence (Proposition 16.2.3(c)) under any weighting of the temperatures.

```
import itertools

import numpy as np
from scipy import stats

flours = ["white", "wholemeal", "rye blend"]
temps = [24, 28, 32, 36]
true_means = np.array([[1880, 2010, 2130, 2190],
                        [1660, 1760, 1810, 1750],
                        [1500, 1540, 1580, 1590]], dtype=float)

a, b, m = 3, 4, 3
data_rng = np.random.default_rng(16_03)
volume = np.round(true_means[:, :, None] + data_rng.normal(scale=55, size=(a, b, m))
                  )

def Jbar(k):
    return np.full((k, k), 1.0 / k)
```

```

def Cen(k):
    return np.eye(k) - Jbar(k)

y = volume.ravel()                                # replicate fastest, then
                                                temperature, then flour
P = {"flour": np.kron(np.kron(Cen(a), Jbar(b)), Jbar(m)),
     "temperature": np.kron(np.kron(Jbar(a), Cen(b)), Jbar(m)),
     "interaction": np.kron(np.kron(Cen(a), Cen(b)), Jbar(m)),
     "error": np.kron(np.kron(np.eye(a), np.eye(b)), Cen(m))}
ss = {k: y @ Pk @ y for k, Pk in P.items()}
df = {k: int(round(np.trace(Pk))) for k, Pk in P.items()}
mse = ss["error"] / df["error"]
for k in ["flour", "temperature", "interaction", "error"]:
    F = ss[k] / df[k] / mse
    tail = f"F = {F:7.2f}  p = {stats.f.sf(F, df[k], df['error']):.2g}" if k != "
    error" else ""
    print(f"{k:12s} df {df[k]:2d}  SS {ss[k]:10.0f}  MS {ss[k] / df[k]:9.0f}  {tail}"
          ")

```

[Run this code online](#)

```

cell = volume.mean(axis=2)
flour_c = {"white vs others": np.array([2.0, -1, -1]), "wholemeal vs rye": np.array
          ([0.0, 1, -1])}
temp_c = {"linear": np.array([-3.0, -1, 1, 3]), "quadratic": np.array([1.0, -1, -1,
          1]),
          "cubic": np.array([-1.0, 3, -3, 1])}
rows = []
for (fn, c), (tn, d) in itertools.product(flour_c.items(), temp_c.items()):
    D = np.outer(c, d)                                # an interaction contrast in
    the cell means
    est = (D * cell).sum()
    ss_c = est ** 2 / ((D ** 2).sum() / m)              # its one-degree-of-freedom
    sum of squares
    rows.append((fn, tn, est, ss_c, ss_c / mse))
    print(f"{fn:17s} x {tn:9s}  estimate {est:8.1f}  SS {ss_c:8.0f}  F {ss_c / mse
    :6.2f}")
print(f"sum of the six: {sum(r[3] for r in rows):.0f}  interaction SS: {ss['
interaction']:.0f}")

```

[Run this code online](#)

```

fmean, tmean, grand = cell.mean(1), cell.mean(0), cell.mean()
s = np.sqrt(mse)
nu = df["error"]
q = stats.studentized_range.ppf(0.95, a, nu)
half_tukey = q * s / np.sqrt(b * m)                  # flour means average b*m
loaves
for i, k in itertools.combinations(range(a), 2):
    d = fmean[i] - fmean[k]
    print(f"{flours[i]:9s} - {flours[k]:9s}: {d:7.1f}  Tukey [{d - half_tukey:7.1f},
    {d + half_tukey:7.1f}]")
t_bonf = stats.t.ppf(1 - 0.05 / (2 * b), nu)          # white - wholemeal at each
temperature

```

```

half_simple = t_bonf * s * np.sqrt(2 / m)
for j in range(b):
    d = cell[0, j] - cell[1, j]
    print(f"white - wholemeal at {temps[j]} C: {d:6.1f} Bonferroni [{d -
        half_simple:6.1f}, {d + half_simple:6.1f}]")
c_scheffe = np.sqrt(df["interaction"] * stats.f.ppf(0.95, df["interaction"], nu))

```

[Run this code online](#)

Warning. It is tempting to drop a nonsignificant interaction and “pool” its sum of squares into the error. With $ab(m-1)$ error degrees of freedom already available this gains little. If the interaction is real, the pooled mean square overestimates σ^2 , and if the decision to pool depends on the data, the nominal levels no longer hold (Section 16.4).

16.3.4 Exercises

16.3.4.1 A. Check your understanding

Exercise A1.

Let $SS_{\text{cells}} = m \sum_{i,j} (\bar{y}_{ij\cdot} - \bar{y}_{\dots})^2$, the between-cells sum of squares of the one-way layout with ab groups. Show that $SS_{\text{cells}} = SS_A + SS_B + SS_{AB}$, and identify each term with a projection.

16.3.4.2 B. Practice

Exercise B1.

Suppose all cell means are equal except μ_{11} , which is larger by δ . Find α_i , β_j and γ_{ij} , and show that the noncentralities of F_A , F_B and F_{AB} are

$$\frac{m\delta^2}{\sigma^2} \cdot \frac{a-1}{ab}, \quad \frac{m\delta^2}{\sigma^2} \cdot \frac{b-1}{ab}, \quad \frac{m\delta^2}{\sigma^2} \cdot \frac{(a-1)(b-1)}{ab}.$$

Which test is most likely to detect a single deviant cell when a and b are large?

Solution. The row means are $\bar{\mu}_{1\cdot} = \mu + \delta/b$ and $\bar{\mu}_{i\cdot} = \mu$ otherwise, and the grand mean is $\mu + \delta/(ab)$. So $\alpha_1 = \delta(a-1)/(ab)$ and $\alpha_i = -\delta/(ab)$ for $i \geq 2$, and

$$\sum_i \alpha_i^2 = \frac{\delta^2}{a^2 b^2} ((a-1)^2 + (a-1)) = \frac{\delta^2(a-1)}{ab^2},$$

so $bm \sum_i \alpha_i^2 / \sigma^2 = m\delta^2(a-1)/(ab\sigma^2)$. By symmetry the noncentrality for B is $m\delta^2(b-1)/(ab\sigma^2)$. For the interaction, $\gamma_{ij} = \delta(g_i - 1/a)(h_j - 1/b)$, where $g_1 = h_1 = 1$ and $g_i = h_j = 0$ for $i, j \geq 2$, so $\sum \gamma_{ij}^2 = \delta^2 \cdot \frac{a-1}{a} \cdot \frac{b-1}{b}$, and the noncentrality is $m\delta^2(a-1)(b-1)/(ab\sigma^2)$. For large a, b the interaction noncentrality approaches $m\delta^2/\sigma^2$, while the main-effect noncentralities approach $m\delta^2/(b\sigma^2)$ and $m\delta^2/(a\sigma^2)$. Almost all of a single deviant cell shows up as interaction. It is, however, spread over $(a-1)(b-1)$ degrees of freedom, and the Scheffé-type contrast aimed at that cell is the one to look at.

Exercise B2.

In the cell-means model Eq. 16.3.1 with normal errors, show that $\bar{y}_{1..}, \dots, \bar{y}_{a..}$ are independent, each with variance $\sigma^2/(bm)$, and jointly independent of SSE, whatever the μ_{ij} . Deduce Tukey intervals for all $\bar{\mu}_{i.} - \bar{\mu}_{k.}$ with exact simultaneous coverage.

Solution. Different rows use disjoint sets of the independent observations, so the row means are independent, and each averages bm observations. The vector of row means is a linear function of $(\mathbf{I}_a \otimes \mathbf{J}_b \otimes \mathbf{J}_m)\mathbf{Y}$, whose range lies in $\mathcal{C}(\mathbf{P}_0 + \mathbf{P}_A) \perp \mathcal{C}(\mathbf{P}_E)$. By Theorem 4.4.4, $(\mathbf{P}_0 + \mathbf{P}_A)\mathbf{Y}$ and $\mathbf{P}_E\mathbf{Y}$ are independent, and $\text{SSE} = \|\mathbf{P}_E\mathbf{Y}\|^2$. The row means have expectations $\bar{\mu}_{i.}$. These are exactly the conditions used in the proof of Theorem 13.3.2, with $g = a$, common variance $\sigma^2/(bm)$ and $\nu = ab(m-1)$. So the intervals $\bar{y}_{i..} - \bar{y}_{k..} \pm q_\alpha(a, ab(m-1))s/\sqrt{bm}$ have simultaneous coverage exactly $1 - \alpha$.

Exercise B3.

Let the levels of A and B be distinct numbers $w_1, \dots, w_a$ and $z_1, \dots, z_b$. Show that the vectors with entries $w_i^r z_j^s$ (repeated over replicates), $0 \leq r < a$, $0 \leq s < b$, span the cell-means space $\mathcal{C}(\mathbf{X})$. Show that the additive model corresponds to the terms with $r = 0$ or $s = 0$. What does the coefficient of wz measure?

16.3.4.3 C. Going deeper

Exercise C1.

Let $\hat{\gamma}_{ij} = \bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}$. Show that among all interaction contrasts, $\text{SS}(\mathbf{d})$ of Eq. 16.3.3 is largest for $d_{ij} \propto \hat{\gamma}_{ij}$, and that the maximum is SS_{AB} . Relate this to Corollary 13.2.3(a). Why does it mean that the F statistic of a contrast chosen after inspecting the interaction plot cannot be compared with $F(1, ab(m-1))$?

Solution. Since (d_{ij}) has zero margins, $\sum d_{ij}\bar{y}_{ij.} = \sum d_{ij}\hat{\gamma}_{ij}$ (Proposition 16.2.1 applied to the table of cell means). By the Cauchy–Schwarz inequality, $(\sum d_{ij}\hat{\gamma}_{ij})^2 \leq \sum d_{ij}^2 \sum \hat{\gamma}_{ij}^2$, with equality iff $\mathbf{d} \propto \hat{\boldsymbol{\gamma}}$. Hence $\text{SS}(\mathbf{d}) \leq m \sum \hat{\gamma}_{ij}^2 = \text{SS}_{AB}$, with equality at $\mathbf{d} = \hat{\boldsymbol{\gamma}}$, which is itself a zero-margin table. In the notation of Corollary 13.2.3(a) this says that the largest single-contrast F ratio is $(a-1)(b-1)F_{AB}$. A contrast chosen for its large value is, in the extreme, this maximizing contrast. Its null distribution is that of $(a-1)(b-1)$ times an $F((a-1)(b-1), ab(m-1))$ variable, not $F(1, ab(m-1))$.

Exercise C2.

For each column j , let $\text{SS}_{A|j} = m \sum_i (\bar{y}_{ij.} - \bar{y}_{.j.})^2$ be the sum of squares for A computed within column j alone. Show that $\sum_j \text{SS}_{A|j} = \text{SS}_A + \text{SS}_{AB}$, and find the projection whose squared length is the left side. What hypothesis does the corresponding F test with $b(a-1)$ numerator degrees of freedom address?